```
write syntax of
1.for loop inside while loop
2.while loop inside for loop
'''

1.
intialization of while
while(condition):
    for itt_var in range(start,end,step):
        statement of for

    statements of while
    increment/decrement of while

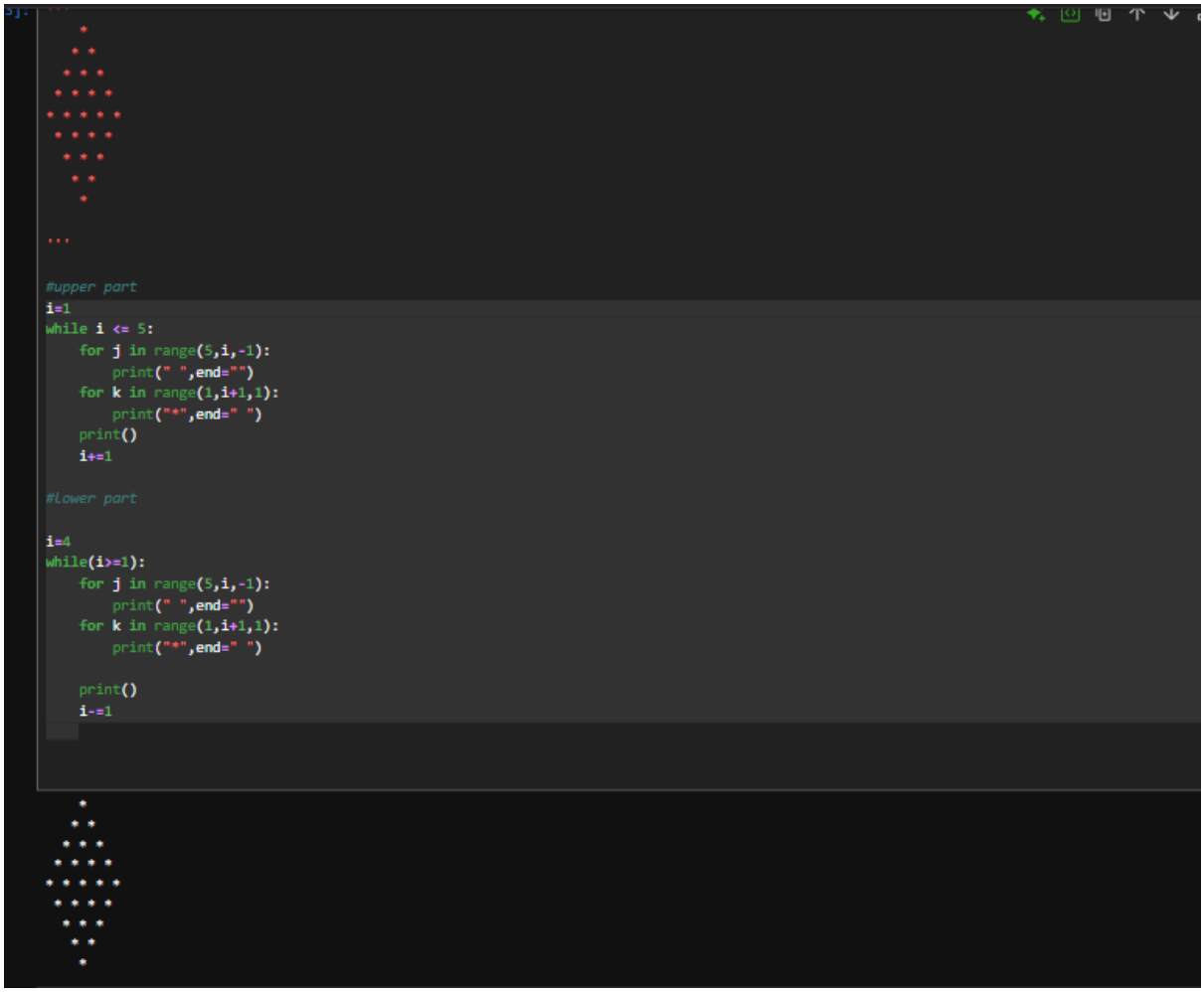

2.
for(start,end,step):
        intialization
        while(condition):
              statement
              increment/decrement
        statement of for
```

[2]:

```
'''
*****
*   *
*   *
*   *
*****
'''

for i in range(1,6):
    j=1
    while(j<=5):
        if(i==1 or i==5 or j==1 or j==5):
            print("*",end=" ")
        else:
            print(" ",end=" ")
        j+=1
    print()
```

```
* * * * *
*       *
*       *
*       *
* * * * *
```


'''
*
* *
* * *
* * * *
* * * * *
* * * *
* * *
* *
*
'''
#upper part
i=1
while i <= 5:
for j in range(5,i,-1):
print(" ",end="")
for k in range(1,i+1,1):
print("*",end=" ")
print()
i+=1
#Lower part
i=4
while(i>=1):
for j in range(5,i,-1):
print(" ",end="")
for k in range(1,i+1,1):
print("*",end=" ")
print()
i-=1
*
* *
* * *
* * * *
* * * * *
* * * *
* * *
* *
*

[4]:
```
'''
---- *
---*   *
--*     *
-*       *
*         *
 *       *
  *     *
   *   *
    *

    '''

#upper part
for i in range(1,6,1):
    j=5
    while(j>i):
        print(" ",end="")
        j-=1
    k=1
    while(k<=i):
        if(i==1 or k==1 or k==i):
            print("*",end=" ")
        else:
            print(" ",end=" ")
        k+=1
    print()

#Lower part
for i in range(1,5,1):
    j=1
    while(j<=i):
        print(" ",end="")
        j+=1
    k=4
    while(k>=i):
        if(i==4 or k==4 or k==i):
            print("*",end=" ")
        else:
            print(" ",end=" ")
        k-=1
    print()
```

```
    *
   * *
  *   *
 *     *
*       *
 *     *
  *   *
   * *
    *
```

[5]:
```
'''
A
A B
C E F
G H I J
K L M N O

'''

ch = 65
for i in range(1,6):
    j=1
    while(j<=i):
        print(chr(ch),end=" ")
        ch+=1
        j+=1
    print()
```

```
A
B C
D E F
G H I J
K L M N O
```

[6]:

```
'''
P Q R S
T U V
W X
Y
'''
ch = 81
for i in range(1,5):
    j=4
    while(j>=i):
        print(chr(ch),end=" ")
        ch+=1
        j-=1
    print()
```

```
Q R S T
U V W
X Y
Z
```